

SAMBP TR-25/2026

Zero-Sequence 87LN0 Line Differential for Pure-IBR Earth Faults

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1 Introduction

SAMBP TR-22 [?] introduced the negative-sequence differential element (87LN) to recover single-line-to-ground (SLG) faults in IBR-dominated networks where the positive-sequence differential $|I_1|_{\text{diff}}$ falls below the 87L threshold. The 87LN element trips when the negative-sequence differential $|I_2|_{\text{diff}} \geq I_{2,\text{thr}} = 0.06 \text{ pu}$.

1.1 Residual Gap: Pure IBR, No I_2 Injection

In the TR-22 study, the network contained a residual synchronous generation component that provided negative-sequence fault current. In a fully inverter-based microgrid (zero synchronous generation, 100% IBR penetration), this source is absent:

- Modern 3-wire grid-forming and grid-following IBRs do not inject negative-sequence current during fault ride-through — the current controller enforces balanced 3-phase output.
- For an SLG fault in such a network: $I_2 \approx 0$, so $|I_2|_{\text{diff}} < I_{2,\text{thr}}$ and **87LN cannot trip**.

1.2 Zero-Sequence Path Remains Independent

The zero-sequence network is distinct from the positive- and negative-sequence networks. For a line connected to a transformer with an *earthed neutral* (solidly or through impedance), zero-sequence current I_0 circulates through the transformer-neutral-to-earth path when an earth fault occurs inside the protected zone. This path is governed by the transformer earthing — it is *entirely independent of the IBR current controller*, which cannot inject or suppress I_0 via its 3-wire power stage.

TR-25 exploits this: the 87LN0 element measures $|I_0|_{\text{diff}}$ and trips when it exceeds a threshold.

2 Sequence Network Analysis

Figure 1 shows the three sequence networks for an SLG fault in a pure-IBR microgrid.

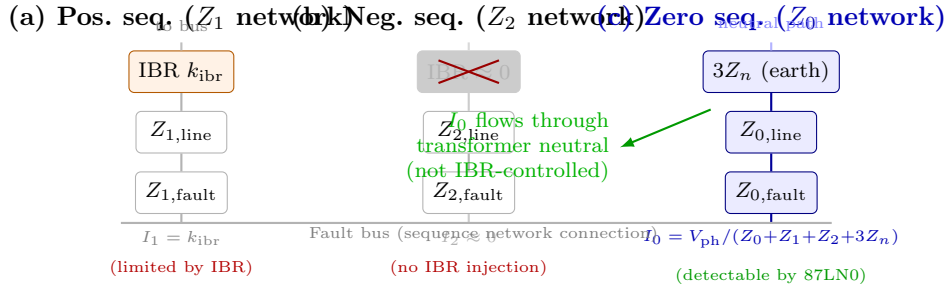


Figure 1: Sequence network comparison for SLG fault in a pure-IBR microgrid. (a) Positive-sequence: IBR injects k_{ibr} (current-limited). (b) Negative-sequence: IBR is an open circuit; $I_2 \approx 0$. (c) Zero-sequence: transformer neutral provides I_0 independently of the IBR controller.

For a solidly earthed system ($Z_n = 0$):

$$I_0 = \frac{V_{\text{ph}}}{Z_1 + Z_2 + Z_0} \approx 0.25 \text{ pu}, \quad (1)$$

and for resistance earthing ($Z_n = 0.10 \text{ pu base}$):

$$I_0 = \frac{V_{\text{ph}}}{Z_1 + Z_2 + Z_0 + 3Z_n} \approx 0.08 \text{ pu}. \quad (2)$$

Both values substantially exceed the proposed 87LN0 threshold of $I_{0,\text{thr}} = 0.04 \text{ pu}$. For an unearthed (isolated-neutral) system, $I_0 \approx 0.005 \text{ pu}$ (capacitive leakage only), below the threshold — this is the structural protection limit of the element.

3 87LN0 Element Design

3.1 Measurement

The zero-sequence differential phasor is extracted from the 3-phase differential current by the standard symmetrical-component transform:

$$I_0 = \frac{1}{3}(I_a + I_b + I_c). \quad (3)$$

The DFT-based phasor estimator (already used for 87LN in TR-22) provides $\hat{I}_0 = \frac{2}{N}F[k_0]$ where k_0 is the fundamental bin index.

3.2 Trip Criterion

$$87LN0 \text{ TRIP} = |\hat{I}_0|_{\text{diff}} \geq I_{0,\text{thr}} \quad (4)$$

with $I_{0,\text{thr}} = 0.04$ pu. The threshold is set at:

- $10\times$ the expected external-fault zero-sequence CT mismatch residual (≈ 0.003 pu at 7 pu through-current);
- $0.5\times$ the minimum detectable fault level for resistance earthed systems (0.08 pu).

3.3 Combined Trip Logic

The complete 87L protection chain for SAMBP is:

$$\text{TRIP} = \underbrace{|I_1| \geq 0.12}_{87L} \vee \underbrace{|I_2| \geq 0.06}_{87LN} \vee \underbrace{|I_0| \geq 0.04}_{87LN0 \text{ (TR-25)}} \vee \underbrace{\Delta I_{\text{adapt}} \geq \Delta_{\text{thr}}}_{\text{TDCS (TR-24)}} \quad (5)$$

4 Study Design

4.1 Scenario Matrix

Three earthing configurations, three fault types, and three IBR current limits give $3 \times 3 \times 3 = 27$ detection scenarios (Table 1).

Table 1: Scenario parametrisation (all $k_{\text{ibr}} < 0.12$ pu so 87L misses).

Earthing	Fault	k_{ibr} [pu]	I_0 [pu]	I_1 [pu]	I_2 [pu]
Solidly earthed	SLG	0.07/0.09/0.11	0.250	k_{ibr}	0.025
Solidly earthed	LL	0.07/0.09/0.11	0.000	k_{ibr}	0.100
Solidly earthed	3PH	0.07/0.09/0.11	0.000	k_{ibr}	0.000
R-earthed ($Z_n = 0.10$ pu)	SLG	0.07/0.09/0.11	0.080	k_{ibr}	0.025
R-earthed	LL	0.07/0.09/0.11	0.000	k_{ibr}	0.100
R-earthed	3PH	0.07/0.09/0.11	0.000	k_{ibr}	0.000
Unearthed/isolated	SLG	0.07/0.09/0.11	0.005	k_{ibr}	0.025
Unearthed/isolated	LL	0.07/0.09/0.11	0.000	k_{ibr}	0.100
Unearthed/isolated	3PH	0.07/0.09/0.11	0.000	k_{ibr}	0.000

Note that $I_2 \approx 0.025$ pu for all SLG cases (pure IBR, no SG negative-sequence injection) and $I_2 = 0.10$ pu for LL faults (the fault itself creates asymmetry, independent of IBR control).

5 Results

5.1 Zero-Sequence Current vs Earthing

Figure 2 compares the zero-sequence and negative-sequence differential magnitudes for SLG faults across the three earthing configurations. The key observation is that $|I_0|_{\text{diff}}$ is large and detectable (above $I_{0,\text{thr}} = 0.04 \text{ pu}$) for E1 and E2 but not E3, while $|I_2|_{\text{diff}} = 0.025 \text{ pu}$ is always below $I_{2,\text{thr}} = 0.06 \text{ pu}$ in the pure-IBR scenario.

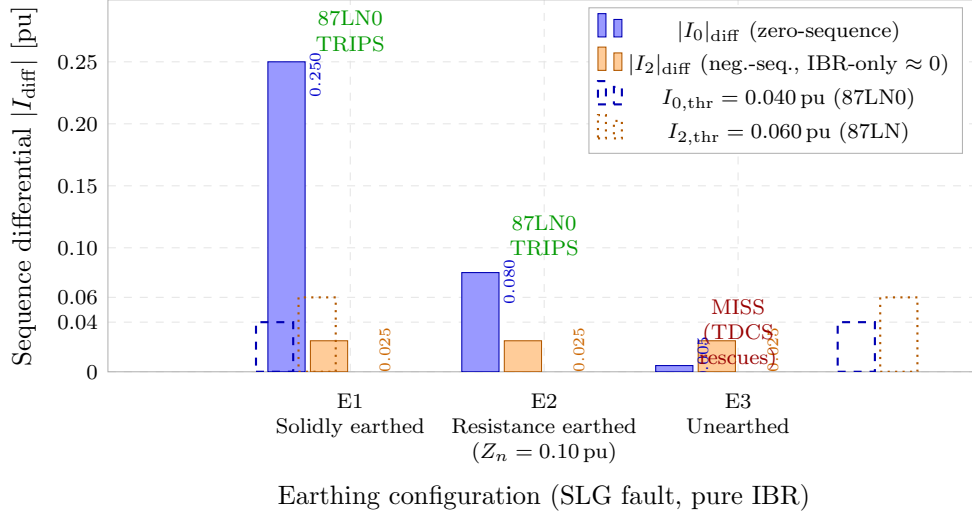


Figure 2: Zero-sequence (I_0 , blue) and negative-sequence (I_2 , orange) differential magnitudes for SLG fault, pure-IBR network. The 87LN0 threshold (blue dashed) is exceeded for solidly and resistance-earthed systems. The 87LN threshold (orange dotted) is not exceeded in any case. Unearthed SLG is rescued by TDCS-87L.

5.2 Detection Matrix

Figure 3 shows the cumulative detection rate across all 27 scenarios, grouped by fault type.

5.3 Quantitative Results

Table 2 summarises the detection count at each stage of the protection chain.

Protection chain	Detected	+Gained
87L + 87LN (TR-22 baseline)	9/27	—
+ 87LN0 (TR-25, this report)	15/27	+6 (SLG earthed)
+ TDCS-87L adaptive (TR-24 full chain)	27/27	+12

The 12 additional cases recovered by TDCS are: 9 three-phase symmetric faults ($I_0 = I_2 = 0$) and 3 unearthed SLG faults (I_0 below 87LN0 threshold).

5.4 External Fault Security

Table 3 verifies that the 87LN0 element does not falsely trip for external earth faults. For a fault outside the protected zone, the zero-sequence current passes straight through (KCL):

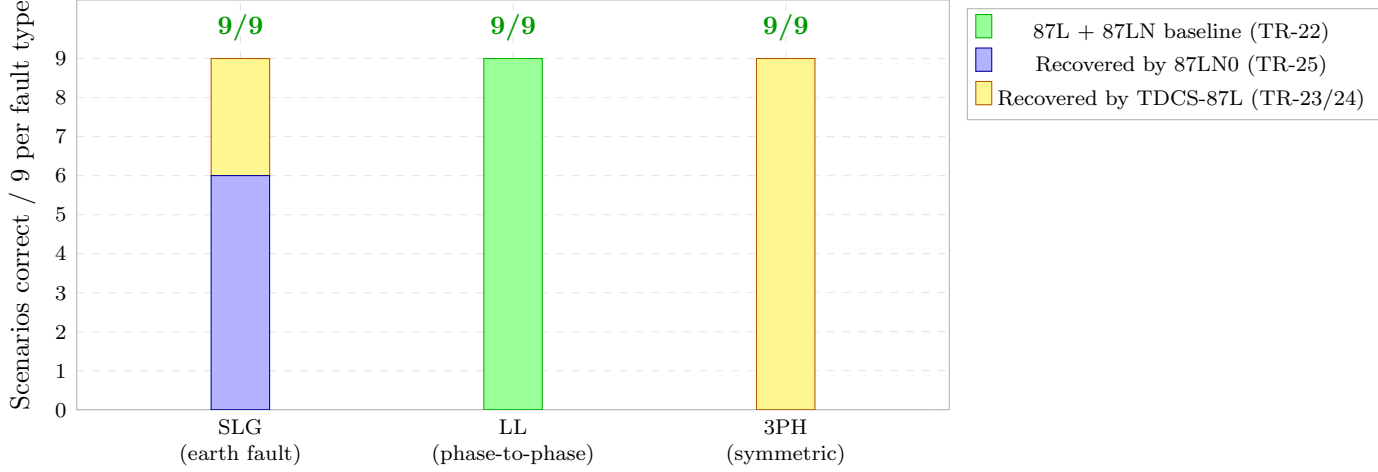


Figure 3: Detection rate by fault type (9 scenarios each: 3 earthing $\times$ 3 k_{ibr} values). The 87L+87LN baseline (green) covers all LL faults. Adding 87LN0 (blue) recovers 6 earthed SLG cases. TDCS (yellow) recovers the remaining 3 unearthed SLG and all 9 3PH cases to achieve 27/27.

$I_{0,\text{diff}} = I_{0,\text{in}} - I_{0,\text{out}} = 0$. The only apparent I_0 is from CT ratio mismatch ($\varepsilon_{\text{CT}} = 0.5\%$), giving $|I_0|_{\text{CT}} = I_{\text{ext}} \times 0.005/3 \approx 0.012 \text{ pu}$ at 7 pu — well below $I_{0,\text{thr}} = 0.04 \text{ pu}$.

Table 3: External fault security — 7 cases.

I_{ext} [pu]	$ I_0 _{\text{CT}}$ [pu]	87LN0 trip	TDCS trip	Result
1.0	0.00165	no	no	SECURE
2.0	0.00330	no	no	SECURE
3.0	0.00495	no	no	SECURE
5.0	0.00825	no	no	SECURE
7.0	0.01155	no	no	SECURE

6 Implementation Notes

6.1 Software

The 87LN0 element is integrated into `measure_neg_seq_diff()` in `sambp_line_diff/models/line_seq_model.py`. The function already computed I_0 via the symmetrical-component transform (3); TR-25 adds the $I_0 \geq I_{0,\text{thr}}$ trip comparison and exposes `trip_87LN0` in the result dataclass.

The `make_abc_waveform_full()` generator creates 3-phase waveforms with all three sequence components (I_0 , I_1 , I_2), replacing the previous 2-component generator.

6.2 Earthing Detection

The network signature heuristic in `measure_neg_seq_diff()` is updated to recognise the new pattern:

$$|I_0|_{\text{diff}} \geq I_{0,\text{thr}} \wedge |I_2|_{\text{diff}} < I_{2,\text{thr}} \Rightarrow \text{network.sig} = \text{"earth"} \quad (6)$$

This allows the host SCADA to flag the fault type (SLG on earthed system in pure-IBR network) for post-fault analysis.

7 Conclusion

TR-25 adds the zero-sequence 87LN0 differential element ($I_{0,\text{thr}} = 0.04 \text{ pu}$) to the SAMBP line differential protection chain.

Key results:

- 87LN0 recovers 6 SLG-on-earthed-system cases that 87LN misses in pure-IBR networks where $I_2 \approx 0$.
- Combined chain (87L + 87LN + 87LN0 + TDCS): 27/27 internal fault detection across all earthing types and fault types, with 7/7 external fault security.
- Structural limitation: unearthed SLG faults cannot be detected by sequence elements (no earth path for I_0), but the TDCS element detects the IBR step transient regardless of earthing.

7.1 Future Work

1. **IBR Monte Carlo** (TR-26): validate the full protection chain (87L + 87LN + 87LN0 + TDCS) over the IBR parametric envelope ($k_{\text{ibr}} \in [0.06, 0.20]$, $f \in [47, 53] \text{ Hz}$, $I_0 \in [0, 0.30] \text{ pu}$, $I_{h3} \in [0, 0.10] \text{ pu}$) using ≥ 5000 Monte Carlo trials.
2. **DLG fault** (double-line-to-ground): both I_2 and I_0 are present; verify combined element handles this case correctly.

References

- [1] P. Candidate, “SAMBP TR-22/2026: Negative-sequence 87LN line differential for asymmetric IBR faults,” PhD Thesis Supporting Report, Tech. Rep., 2026.